

StockWise:

AI-Powered Decision System for Inventory Reordering and Waste Reduction

Business Proposal

1. Executive Summary

StockWise is a Malaysia-first inventory decision support application for small cafes, food service operators, and micro or SME retailers that still manage stock through spreadsheets, WhatsApp notes, or manual judgement. The current product is a working MVP, not a finished enterprise platform. It helps owners upload a CSV or enter item records manually, then turns those records into ranked actions such as RESTOCK_NOW, BUY_LESS, DELAY_PURCHASE, and MONITOR_CLOSELY.

The main business problem is simple: small operators often lose cash in two directions at once. If they buy too much, perishable goods become waste. If they buy too little, they risk stockouts and disrupted sales. StockWise addresses this through deterministic scoring first, then AI-assisted explanation second. The system calculates reorder urgency, waste risk, days of cover, estimated waste exposure, and simulation outcomes using explicit formulas. Z.AI GLM is used to translate these results into owner-readable decision briefs, item explanations, and AI Advisor responses, while the numeric source of truth remains the backend scoring engine.

The initial target market is not all Southeast Asia. A realistic entry point is Malaysian independent cafes and small F&B operators, especially in Klang Valley, where the owner can test the tool with existing CSVs before paying. The proposed business model is a low-cost SaaS subscription with a free demo tier, a Starter plan for small shops, a Pro plan for historical CSVs and AI Advisor usage, and optional onboarding support. The 18-month goal is to validate retention and measurable decision usefulness with a modest paid pilot before attempting wider expansion.

2. Product Overview / Solution

Key Features:

- CSV upload and manual entry: Owners can import existing inventory files or key in daily stock observations without needing ERP training.
- Unstructured Data Entry: Allows owners to paste messy supplier messages or quick notes (e.g., WhatsApp texts) to be converted into structured records.
- Ranked inventory actions: The dashboard shows which items need restocking, which items should be bought less, which purchases can be delayed, and which items need monitoring.
- Deterministic scoring engine: Reorder urgency and waste risk are calculated from current stock, usage, lead time, price, seasonality, and waste signal inputs.
- Records review and edit flow: Owners can inspect latest item snapshots and the source observation history behind an analysis before acting on recommendations.
- Simulation engine: Owners can test a proposed reorder quantity and see cash outlay, coverage days, inventory value, waste exposure, risk change, and updated recommendation.
- AI Decision Brief and AI Advisor: GLM-generated summaries turn the analysis into plain-language operating guidance, with validation and deterministic fallback when AI output is invalid.
- Export Analysis and Business Value Snapshot: The export page presents estimated monthly opportunities in RM without claiming realized savings or ROI.

Technology Used:

- FastAPI backend for CSV ingestion, manual analysis, records, simulation, explanation, AI Advisor, and decision brief endpoints.
- React / Next.js frontend with dashboard, records review, simulation, explanation, AI Advisor, and export/report pages.
- Supabase persistence for source observations, inventory records, import batches, user-owned snapshots, and analysis history.
- OpenAI-compatible Z.AI / ILMU GLM integration for structured explanation, decision brief, chat, and simulation verdict payloads.

- Parser and fallback safety layer that rejects invalid JSON, unsupported claims, mismatched item references, and unsafe AI outputs before showing them to users.

3. Market Research & Market Dynamics

3.1 Problem Landscape

Small F&B operators in Malaysia face a practical inventory problem rather than a lack of dashboards. Many owners already know their business from experience, but they do not always have time to convert daily stock records into reorder timing, waste risk, and cash trade-offs. Existing ERP and POS inventory modules can be too costly, too complex, or too heavy for a micro business. Spreadsheets are flexible but do not automatically explain which item should be bought today, which purchase should be delayed, and what risk changes when a reorder quantity changes.

The issue is commercially meaningful. DOSM reports that Malaysia's F&B services sector had 136,453 establishments and RM99.0 billion gross output in 2022. Academic Malaysia hospitality and food service case studies also found food waste could represent a substantial economic loss, with an average around 23% of food purchase value in the studied operations. StockWise does not claim it can remove that loss. It targets the more realistic gap: giving small operators a low-friction way to see avoidable waste and stockout risks earlier.

3.2 Market Niche

StockWise starts with Malaysian independent cafes, beverage shops, bakeries, small restaurants, and small retailers with perishable or semi-perishable stock. These businesses are large enough to feel inventory pain, but often too small to justify a full ERP rollout. The first useful user is an owner-manager or operations lead who can export a CSV, estimate price and usage, and wants a fast daily or weekly buying plan.

The first geography should be Klang Valley and nearby university or SME networks, not nationwide expansion on day one. This is realistic for a student team: it allows direct onboarding, observation of real workflows, and fast correction of product assumptions.

3.3 Market Sizing

Market Tier	Definition	Conservative Estimate
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TAM	Malaysia MSME and F&B operating base. DOSM reported RM652.4b MSME value added in 2024, and F&B services alone had RM99.0b gross output in 2022.	Large national opportunity; not directly capturable by a student MVP.
SAM	Malaysia F&B and small retail operators that already use spreadsheets, CSV exports, or POS data and can try a lightweight web tool.	Approx. 10,000-25,000 reachable tech-ready operators over time.
SOM	Realistic 18-month reachable pilot market through Klang Valley, student networks, SME communities, and direct outreach.	50-100 paying organizations if retention and onboarding prove strong.

3.4 Market Dynamics & Placement

- Cost sensitivity: Micro and small operators need pricing that feels closer to a utility bill than enterprise software.
- Digitalisation gap: Research on Malaysian SMEs notes that cloud adoption still faces resource constraints, capability gaps, and risk aversion, so onboarding must be simple.
- AI curiosity with trust concerns: Owners may like AI summaries, but they still need visible calculations and editable records before trusting reorder advice.
- Waste and cash pressure: F&B operators feel both overstock waste and shortage risk, making inventory decisions a weekly operational pain rather than a theoretical analytics problem.
- Student-team advantage and constraint: The team can move quickly and work closely with early users, but must avoid pretending to have enterprise support capacity before it exists.

4. Value Proposition & Competitive Advantage

StockWise offers a practical middle ground between spreadsheets and full inventory ERP. The owner does not need to migrate their entire operation. They can upload a CSV or enter a few stock records, then receive a ranked action plan, editable records, simulations, and AI-assisted explanations.

The competitive advantage is not that StockWise has the most features. It is that the product is built around a safer decision architecture: deterministic calculations produce the numbers, while

GLM turns those numbers into readable operating guidance. This reduces the risk of AI inventing unsupported financial claims or hidden calculations. The app also keeps Records Review visible, so owners can correct inputs before relying on a recommendation.

For hackathon and early pilot use, StockWise can win by being narrow, explainable, and affordable. It should be judged as a student MVP with a clear route to validation, not as a fully mature ERP replacement.

5. Competitor Analysis (Porter's Five Forces)

Force	Intensity	Assessment
Competitive Rivalry	Medium	Large POS, ERP, accounting, and inventory systems already exist. StockWise competes by being lighter, cheaper, and focused on decision support for small operators rather than full operations management.
Threat of New Entrants	Medium	Simple AI wrappers are easy to build, but a reliable system needs validation, deterministic scoring, fallback behavior, and user trust from real pilots.
Bargaining Power of Buyers	High	SME buyers are price-sensitive and can stay with spreadsheets if setup feels difficult or value is unclear.
Bargaining Power of Suppliers	Medium	Cloud hosting, database, and AI model providers can affect costs. The product reduces dependency by keeping core scoring deterministic and using AI only for explanation.
Threat of Substitutes	High	Google Sheets, Excel, POS exports, manual judgement, and staff experience remain strong substitutes, especially for very small shops.

Competitor Comparison Table:

Competitor / Substitute	Strength	Weakness	StockWise Advantage
Excel / Google Sheets	Almost free, familiar, flexible.	No automatic risk ranking, simulation, or guided decision brief.	Uses existing CSV habits but adds scoring and explanation.
POS inventory modules	Already connected to sales workflow.	May be locked to vendor ecosystem and can be too heavy for simple pilots.	Can start from exported CSV without replacing POS.
Full ERP systems	Powerful, integrated, scalable.	Expensive and complex for micro operators.	Focused MVP for one job: better buying decisions.
Generic AI chatbots	Easy to ask questions in plain language.	May hallucinate numbers and lacks inventory data validation.	AI is bounded by deterministic metrics and parser checks.

6. Business Model Canvas

6.1. Customer Segmentation

- Primary: Malaysian independent cafes, bakeries, beverage shops, and small restaurants with recurring stock decisions.
- Secondary: Small retailers with perishable or fast-moving stock, such as mini marts or specialty food sellers.
- Early adopter profile: Owner-manager who already tracks stock in CSV/spreadsheets and is willing to test a low-cost tool before investing in ERP.

6.2. Revenue Stream

Plan	Indicative Price	Included Value
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Free / Demo	RM0	Basic features such as uploading datasets, exporting records, basic dashboards, AI decision brief, simulation
Starter	RM29-RM49 / month	AI text extraction for input, AI explanation and other basic features
Pro	RM79-RM129 / month	AI advisor and more token usages for AI features implemented on Starter Plan
Setup Help	RM100-RM300 one-time	Optional onboarding, CSV cleanup, and first dashboard walkthrough for operators who need help.

6.3. Cost Structure

Cost Driver	Cost Type	Conservative Notes
Hosting and database	Fixed / usage-based	Vercel, Render, and Supabase can start low but will rise with users and stored history.
AI provider usage	Variable	GLM calls should be capped per plan because chat and briefs create token costs.
Onboarding and support	Time cost	Highest near-term cost for a student team; direct support is useful for learning but cannot scale manually forever.
Product development	Fixed time cost	Student development effort, QA, design, documentation, and bug fixing.
Marketing	Variable	Initial outreach should use student networks, cafe visits, SME

		communities, and referrals before paid ads.
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6.4. Key Partners

Partner Type	Potential Role
University and hackathon network	Pilot introductions, judging feedback, credibility for first conversations.
Local cafes / F&B operators	Friendly pilots, real CSV formats, workflow feedback, testimonials if results are useful.
SME associations and entrepreneurship clubs	Distribution channel for workshops and onboarding sessions.
Cloud and AI providers	Infrastructure for hosting, data persistence, and GLM explanation features.

6.5 Risk Assessment

Risk Type	Impact	Likelihood	Mitigation
Insufficient real-user validation	High	Medium	Run 3-5 friendly pilots before claiming market readiness.
AI hallucination or unsafe advice	High	Medium	Keep deterministic metrics as source of truth; validate AI JSON; use fallback responses.
SMEs unwilling to pay	High	Medium	Keep a free demo and low Starter price; prove usefulness before pushing Pro.
Onboarding burden	Medium	High	Provide CSV templates, guided fields, and optional paid setup help.

Data privacy and trust	High	Medium	Use authenticated user-owned data, avoid hardcoded secrets, and explain data handling clearly.
Student-team capacity	Medium	High	Limit early scope, avoid custom integrations until retention is proven.

7. Implementation & Financials

7.1 Go-to-Market Strategy

The go-to-market plan should be pilot-led rather than advertising-led. The first objective is not thousands of users; it is evidence that real Malaysian operators can upload or enter inventory records, understand the recommendations, and return to the product in the next stock review cycle. Early acquisition should come from direct cafe outreach, university entrepreneurship networks, SME workshops, and referrals from pilot users.

StockWise should start with Klang Valley because it is accessible for in-person onboarding and has a dense SME base. The message should be practical: upload your stock sheet, see what to buy today, what to buy less, and what can wait. The first case studies should focus on time saved, decision clarity, and estimated opportunity rather than profit promises.

7.2 Implementation Plan

Phase	Action	Timeline	Target
MVP Stabilisation	Fix final deployment gaps, validate core upload, dashboard, records, simulation, AI fallback, and export flows.	Month 1-2	3-5 friendly pilots.
Pilot Learning	Observe CSV formats, onboarding friction, confusing fields, and whether owners act on recommendations.	Month 3-6	20-30 active users, first testimonials if earned.

Paid Beta	Introduce Starter and Pro pricing, usage caps, basic support playbook, and monthly retention tracking.	Month 7-12	50-100 paying or trial organizations.
Focused Expansion	Improve integrations only after repeated demand; expand beyond Klang Valley if retention supports it.	Month 13-18	100-200 active organizations, sustainable support process.

7.3 Revenue Projections (Conservative 3-Year)

These projections are planning estimates for a student-built SaaS pilot. They are not commitments about revenue, ROI, profit, or market capture.

Metric	Year 1	Year 2	Year 3
Active paying organizations by year end	60	220	600
Average subscription revenue per account	RM55 / month	RM75 / month	RM95 / month
Subscription revenue estimate	RM39,600	RM198,000	RM684,000
Setup / onboarding revenue estimate	RM6,000	RM24,000	RM48,000
Total revenue estimate	RM45,600	RM222,000	RM732,000
Estimated operating cost	RM55,000	RM165,000	RM455,000
Net position before founder salary	RM(9,400)	RM57,000	RM277,000

7.4 Key Financial Milestones

- Month 2: Product can support a small live pilot with clear fallback behavior and no critical upload/dashboard blockers.
- Month 6: At least 20-30 active pilot users or trial organizations, with evidence of repeat use across inventory review cycles.
- Month 12: 50-100 paying or serious trial organizations, with pricing validated against support cost and AI usage cost.
- Month 18: Expand only if retention, onboarding, and support load show the product can grow beyond founder-led assistance.

Current MVP Limitations

Limitation	Honest Status
Enterprise readiness	StockWise is not yet a full ERP, POS, procurement, or accounting replacement.
Testing coverage	Backend and frontend unit/API tests exist, but full browser E2E and load testing remain limited.
Deployment evidence	Live deployment still needs final verification, including Supabase snapshot reads after backend redeploy.
AI dependency	AI is useful for explanation but must remain bounded by deterministic scoring, parser checks, and fallback behavior.
Market proof	No traction should be invented. The next milestone is pilot evidence from real operators.

Sources and Evidence Used

Product details are based on the project documentation in docs/project-requirements.md, docs/architecture-and-coding-design.md, docs/frontend-pages-and-fields.md, docs/project-status.md, and docs/testing-analysis-documentation.md.

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